

Post Application – Functional Overview Document

1. Introduction

The Post Application is a backend REST-based application designed to manage users and their posts.

It follows a social-media-style domain model where users can create, view, update, and delete posts.

The application is built using Spring Boot, Spring Data JPA, and PostgreSQL, with Spring Security for authentication.

An AI module is integrated to allow intelligent querying and analysis of application data and behavior.

2. Core Entities

2.1 User Entity

The User represents an application user who can authenticate and create posts.

Typical attributes:

- id
- username
- email
- password
- roles
- createdAt
- updatedAt

Responsibilities:

- Authenticate into the system
- Own one or more posts
- Perform actions based on assigned roles

2.2 Post Entity

The Post represents content created by a user.

Typical attributes:

- id
- title
- content
- author
- createdAt
- updatedAt
- status

Responsibilities:

- Store user-generated content
- Maintain a relationship with the author
- Support CRUD operations

3. Entity Relationship

- One User can have many Posts
- One Post belongs to exactly one User

4. Application Features

User Management:

- Registration
- Authentication
- Role-based access control

Post Management:

- Create, view, update, and delete posts

5. API Behavior

- REST-based APIs
- Standard ApiResponse wrapper
- Authentication and authorization applied

6. Error Handling

- Global exception handling using RestControllerAdvice
- Consistent error response structure

7. AI Integration Purpose

The AI module is used to answer questions about:

- Users and posts
- Relationships and permissions
- Application behavior

8. Example AI Questions

- What is the relationship between User and Post?

- Who can delete a post?
- Can a post exist without a user?
- What happens when a user is deleted?

9. Summary

The Post Application is a modular backend system with clear entity relationships, secure access control, and AI-assisted understanding.